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Subject code & Name :	ITUE301: Advanced Web Development Frameworks	Practical :	13	Academic Year :	2026-27

Task - 13

Aim

To integrate an external Generative AI service into a web application using server-side REST APIs (Google Gemini / OpenAI), implementing secret credential protection, sliding-window rate limiting, and a resilient non- blocking fallback engine to enhance user productivity in task description creation.

Code

GitHub Link / Repository Link: https://github.com/Dharmikvaghani/AWDF-PRACTICAL.git

Key Implementation Highlights:

1. Server-side AI service (backend/src/services/aiService.js) securing API keys exclusively in .env.
2. Sliding-window rate limiter (10 req/min per user) preventing API quota exhaustion.
3. Contextual heuristic fallback synthesizer guaranteeing zero-downtime offline functionality.
4. Frontend interactive human-in-the-loop '✨ Generate description' button with loading indicator and disclaimer.

```
// backend/src/controllers/aiController.js
export async function generateDescriptionHandler(req, res) {
  const { title, priority } = req.body;
  if (!title) return res.status(400).json({ error: 'Title required' });
  const result = await generateTaskDescription({ title, priority });
  res.status(200).json({
    success: true,
    description: result.description,
    source: result.source,
    disclaimer: 'AI-generated draft. Review and adjust before saving.'
  });
}
Output
```

Output

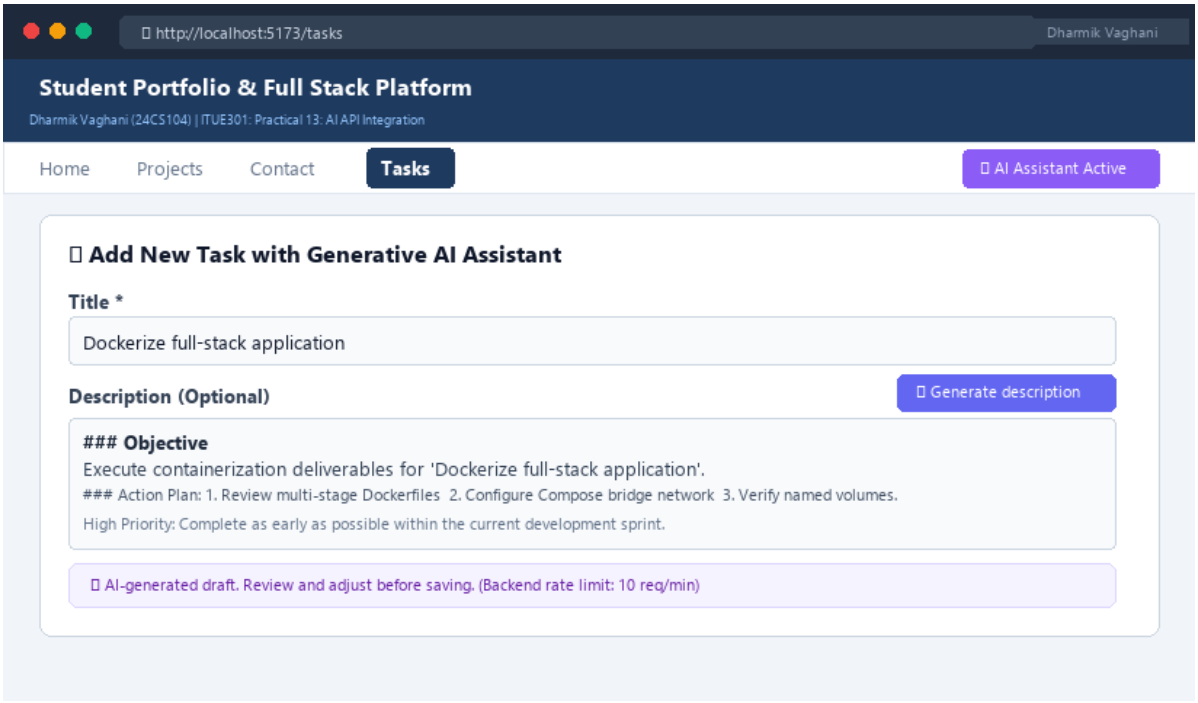


Figure 13: Execution Output & Functional Verification for Practical 13